

Open Science Assignment

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Theoretical hypothesis

My theoretical hypothesis is that **Satyajit Ray** (from Kolkata, India) and **Akira Kurosawa** (from Japan) are both great directors, both won the *academy award for lifetime achievement*. Because they are both great, the *movies they directed are equally good*.

Dependent Variables to be measured

- The dependent variables to be measured are the **IMDB ratings** (scores), **# Users rated each movie**.
- First IMDB search will be used separately for the two legendary directors separately to get all the hits.
- Then the search results will be sorted based on the release date, and 29 most recent full movies (excluding documentaries / TV series) will be used.
- So in my case, I will use the 29 last movies **Satyajit Ray** and **Akira Kurosawa** directed in from today (excluding documentaries / TV series), the moment I did the IMDB search.
- The following table shows the data collected for **Satyajit Ray** movies.

Movie	Rating (Out of 10)	#Users Rated	Release Year
The Stranger (Agantuk)	8.1	1760	1991
Shakha Prosakha	7.6	453	1990
Ganashatru	7.3	662	1989
Ghare Baire	7.7	812	1984
Hirak Rajar Deshe	8.8	1387	1980
Jai Baba Felunath	7.9	1086	1979
Shatranj Ke Khilari	7.8	2370	1977
Jana Aranya	8.3	887	1976
Sonar Kella	8.5	1308	1974
Distant Thunder (Ashani Sanket)	8.2	908	1973
Company Limited (Seemabaddha)	8.0	782	1971
Pratidwandi	8.2	1051	1970
Days and Nights in the Forest (Aranyer Din Ratri)	8.3	1720	1970
Goopy Gayen Bagha Bayen	8.8	1495	1969

Movie	Rating (Out of 10)	#Users Rated	Release Year
Chiriyakhana	7.2	477	1967
Nayak	8.3	1974	1966
Mahapurush	7.3	719	1965
The Coward (Kapurush)	7.8	858	1965
Charulata	8.3	3597	1964
Mahanagar	8.3	2275	1963
Abhijaan	8.0	781	1962
Kanchenjunga	8.0	706	1962
Teen Kanya	8.2	991	1961
Devi	8.0	1407	1960
The World of Apu (Apu Sansar)	8.2	8058	1959
The Music Room (Jalshaghar)	8.1	3872	1958
Paras-Pathar	7.8	723	1958
Aparajito	8.2	7880	1956
Pather Panchali	8.4	15799	1955

- The following table shows the data collected for **Akira Kurosawa** movies.

Movie	Rating (Out of 10)	#Users Rated	Release Year
Maadadayo	7.4	4035	1993
Rhapsody in August	7.3	5131	1991
Dreams	7.8	19373	1990
Ran	8.2	84277	1985
Kagemusha	8.0	25284	1980
Dersu Uzala	8.3	18898	1975
Dodes'ka-den	7.5	4839	1970
Red Beard	8.3	12295	1965
High and Low	8.4	19989	1963
Sanjuro	8.2	22296	1962
Yojimbo	8.3	80906	1961
The Bad Sleep Well	8.1	8082	1960

Movie	Rating (Out of 10)	#Users Rated	Release Year
The Hidden Fortress	8.1	25980	1958
The Lower Depths	7.5	3776	1957
Throne of Blood	8.1	34723	1957
I Live in Fear	7.4	3090	1955
Seven Samurai	8.7	247406	1954
Ikiru	8.3	46692	1952
The Idiot	7.4	3533	1951
Rashomon	8.3	112668	1950
Scandal	7.4	2580	1950
Stray Dog	7.9	11789	1949
The Quiet Duel	7.5	2131	1949
Drunken Angel	7.8	7422	1948
One Wonderful Sunday	7.3	1988	1947
Waga seishun ni kuinashi	7.2	2158	1946
Asu o tsukuru hitobito	6.6	119	1946
The Men Who Tread on the Tiger's Tail	6.8	2567	1945
Zoku Sugata Sanshirô	6.2	1419	1945
Zoku Sugata Sanshirô	5.8	1229	1944

Justify the sample size

- I predict **no difference**, and thus I'll do a **power analysis** for an **equivalence test**. I want to be pretty sure I can reject my **smallest effect size** of interest, so I'll design a study with **84% power**. For this educational assignment, we do not collect a huge amount of data.
- As long as I can exclude a **large effect** (Cohen's $d = 0.8$ or larger) I will be happy for this assignment.
- The **power analysis** estimates that the **sample size** we need to show the difference between the ratings for movies directed by **Satyajit Ray** and **Akira Kurosawa** is smaller than **Cohen's $d = 0.8$** (assuming the **true effect size** is 0, and with an α of 0.05, when I aim for **84% power**) is 29 movie ratings from **Satyajit Ray**, and 29 movie ratings from **Akira Kurosawa**, as can be seen from the following R code and the figures.
- The α -level I found acceptable is 0.05.
- I performed a **two-sided test**.
- I used **84% power** for this study.
- The effect size expected is $0.78948 < 0.8$, as shown below.

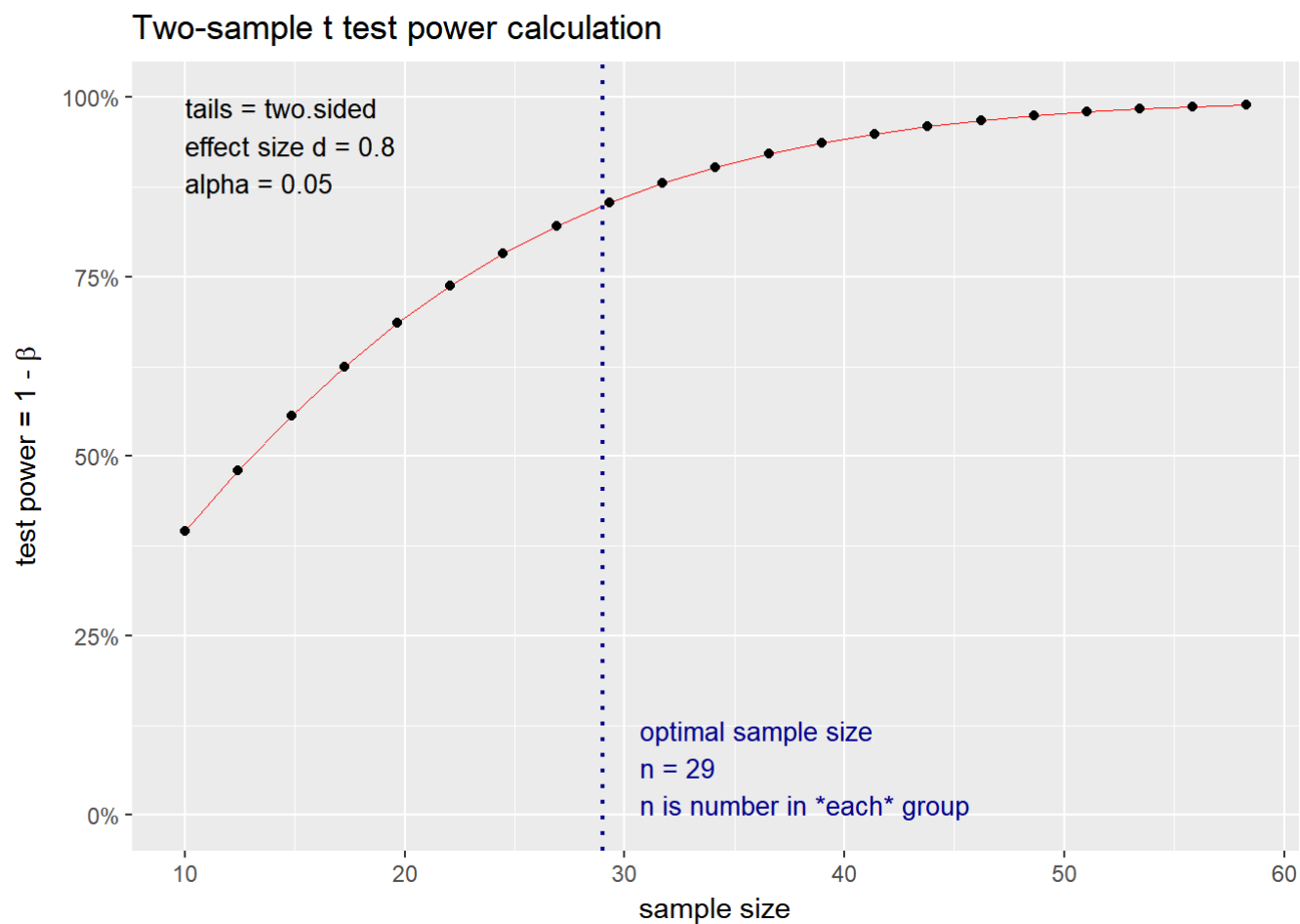
- Given that **Satyajit Ray** has total 29 full movies directed, we can only collect 29 observations for both the directors.

Results

```
library(pwr)
p.t.two <- pwr.t.test(d=0.8,power=0.84,sig.level=0.05,type="two.sample",alternative="two.sided")
print(p.t.two)
```

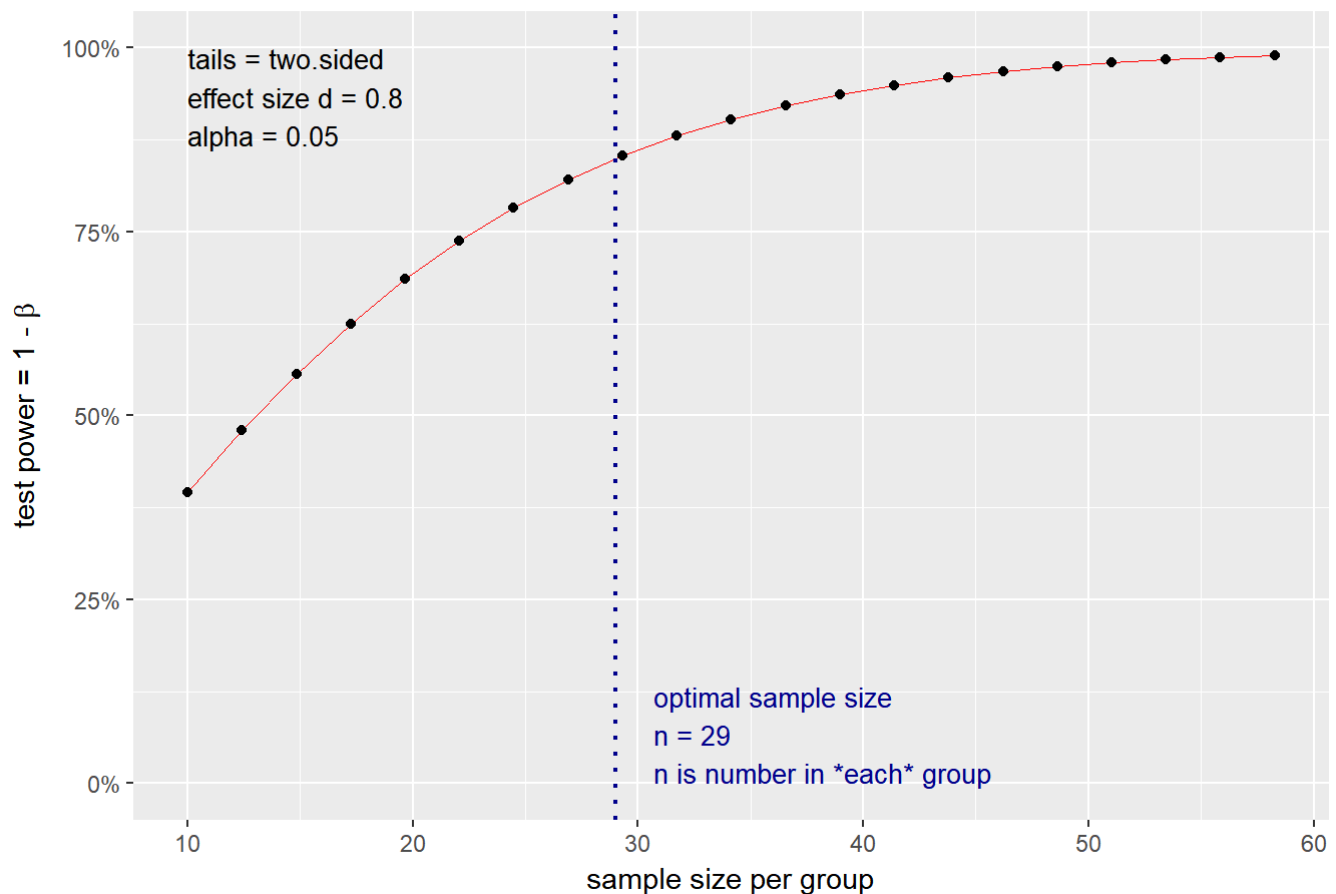
```
##
##      Two-sample t test power calculation
##
##              n = 28.27101
##              d = 0.8
##      sig.level = 0.05
##      power = 0.84
##      alternative = two.sided
##
## NOTE: n is number in *each* group
```

```
plot(p.t.two)
```



```
plot(p.t.two, xlab="sample size per group")
```

Two-sample t test power calculation



```
pwr.t.test(n=29,power=0.84,sig.level=0.05,type="two.sample",alternative="two.sided")
```

```
##
##    Two-sample t test power calculation
##
##          n = 29
##          d = 0.78948
##    sig.level = 0.05
##          power = 0.84
##    alternative = two.sided
##
## NOTE: n is number in *each* group
```

- As can be seen from above, the sample size required to obtain 84% power is 29.

Specify the statistical test to conduct

- We need to translate our **theoretical hypothesis** to a **statistical hypothesis**.
- Let's calculate the (90%) **CI** around the **effect size**.
- When the 90% **CI** falls below, and excludes a **Cohen's d** of 0.8, we can consider the ratings of the movies directed by **Satyajit Ray** and **Akira Kurosawa** as equivalent.

```
# NHST two-sided t-test
# 1. with t.test
t.test(dfs[,2], dfa[,2], alternative='two.sided', conf.level=0.9)
```

```
##
## Welch Two Sample t-test
##
## data: dfs[, 2] and dfa[, 2]
## t = 2.6853, df = 46.646, p-value = 0.01
## alternative hypothesis: true difference in means is not equal to 0
## 90 percent confidence interval:
## 0.1444567 0.6258881
## sample estimates:
## mean of x mean of y
## 8.055172 7.670000
```

```
# 2. with formula
n1 <- length(dfs[,2])
n2 <- length(dfa[,2])
mu1 <- mean(dfs[,2])
mu2 <- mean(dfa[,2])
sd1 <- sd(dfs[,2])
sd2 <- sd(dfa[,2])
df <- n1 + n2 - 2
s.pooled <- sqrt(((n1-1)*sd1^2 + (n2-1)*sd2^2)/df)
t.stat <- (mu1 - mu2 - 0) / (s.pooled * sqrt(1/n1+1/n2))
p.value <- pt(t.stat, df, lower.tail=FALSE)
print(paste0('t.stat=', t.stat, ' df=', df, ' p.value=', p.value,
             ' CI=(', (mu1-mu2)-qt(0.95, df)*s.pooled*sqrt(1/n1+1/n2), ',', (mu1-
mu2)+qt(0.95,df)*s.pooled*sqrt(1/n1+1/n2), ')'))
```

```
## [1] "t.stat=2.66216514176719 df=57 p.value=0.00503579567833758 CI=(0.143256766714653,0.627088
060871554)"
```

```
d <- abs(mu1 - mu2) / s.pooled # choen's d_s
print(paste('Observed effect size (Cohen\'s d_s) = ', d))
```

```
## [1] "Observed effect size (Cohen's d_s) = 0.693268347374739"
```

```
#TOST Equivalence Test
#Low equivalence bound (Cohen's d) -0.8
#high equivalence bound (Cohen's d) 0.8

delta.L <- -0.8
delta.U <- 0.8
#One-Sided Test 1
t.L <- (mu1 - mu2 - delta.L) / (s.pooled * sqrt(1/n1+1/n2))
p.L <- pt(t.L, df, lower.tail=FALSE)
#One-Sided Test 2
t.U <- (mu1 - mu2 - delta.U) / (s.pooled * sqrt(1/n1+1/n2))
p.U <- pt(t.U, df, lower.tail=FALSE)

t.stat <- ifelse(abs(t.L) < abs(t.U), t.L, t.U)
p.value <- max(p.L, p.U)
print(paste0('t.L=', t.L, ' df=', df, ' p.L=', p.L))
```

```
## [1] "t.L=8.19146069136423 df=57 p.L=1.6537791186325e-11"
```

```
print(paste0('t.U=', t.U, ' df=', df, ' p.U=', p.U))
```

```
## [1] "t.U=-2.86713040782984 df=57 p.U=0.997102267946586"
```

```
print(paste0('t.stat=', t.stat, ' df=', df, ' p.value=', p.value))
```

```
## [1] "t.stat=-2.86713040782984 df=57 p.value=0.997102267946586"
```

```
library(compute.es)
tes(t=(mu1 - mu2 - 0) / (s.pooled * sqrt(1/n1+1/n2)), n.1=n1, n.2=n2, level=90)
```

```

## Mean Differences ES:
##
## d [ 90 %CI] = 0.69 [ 0.24 , 1.14 ]
## var(d) = 0.07
## p-value(d) = 0.01
## U3(d) = 75.59 %
## CLES(d) = 68.8 %
## Cliff's Delta = 0.38
##
## g [ 90 %CI] = 0.68 [ 0.24 , 1.13 ]
## var(g) = 0.07
## p-value(g) = 0.01
## U3(g) = 75.3 %
## CLES(g) = 68.57 %
##
## Correlation ES:
##
## r [ 90 %CI] = 0.33 [ 0.12 , 0.51 ]
## var(r) = 0.01
## p-value(r) = 0.01
##
## z [ 90 %CI] = 0.35 [ 0.12 , 0.57 ]
## var(z) = 0.02
## p-value(z) = 0.01
##
## Odds Ratio ES:
##
## OR [ 90 %CI] = 3.52 [ 1.56 , 7.93 ]
## p-value(OR) = 0.01
##
## Log OR [ 90 %CI] = 1.26 [ 0.44 , 2.07 ]
## var(lOR) = 0.24
## p-value(Log OR) = 0.01
##
## Other:
##
## NNT = 4.15
## Total N = 59

```

- As can be seen from the **NHST** test above that the effects are **statistically significant**, since 90% **confidence interval** around the effect size does not contain 0.
- Also, the **TOST** procedure results shown above indicates that the observed effect size $d = 0.69$ was **not significantly** within the **equivalent bounds** of $d = -0.8$ and $d = 0.8$, $t(29) = -2.86$, $p = 0.997$.
- Also, the 90% **CI** (0.24, 1.14) includes a **Cohen's d** of 0.8, hence, we can consider the ratings of the movies directed by **Satyajit Ray** and **Akira Kurosawa** as **not equivalent**.
- Hence, the effect is **statistically significant**, but not **statistically equivalent**.
- **Supporting the alternative with Bayes Factors**: As can be seen from the following results, the **alternative hypothesis** has become 50.17844 times more likely than the **null hypothesis** (as **poterior belief**), with **small prior belief on effect size**.

#This code is made by Jeff Rouder: <http://jeffrouder.blogspot.nl/2016/01/what-priors-should-i-use-part-i.html>

#This code is for a one-sided t-test, testing a difference against 0.

N <- 29 # sample size

dz <- 0.693268347374739 # Cohen's dz effect size observed in the dependent t-test or one-sided t-test

dz_prior <- 0.2 #Enter effect size dz for the prior

sd_prior <- 0.2 #Enter sd of the effect sizes of the prior - the higher, the wider the prior is

lo <- -Inf #lower bound of support (e.g., set to 0 if effects < 0 is not possible)

hi <- Inf #upper bound of support

#specify prior

altDens=function(delta)

*dnorm(delta,dz_prior,sd_prior)*as.integer(delta>lo)*as.integer(delta<hi)*

#Normalize alternative density in case user does not,

K=1/integrate(altDens,lower=lo,upper=hi)\$value

*f=function(delta) K*altDens(delta)*

delta=seq(-3,3,.01)

#Plot Alternative as a density and Null as a point arrow

#png(file=paste('prior.png'),width=6000,height=4000, res = 1000)

maxAlt=max(f(delta))

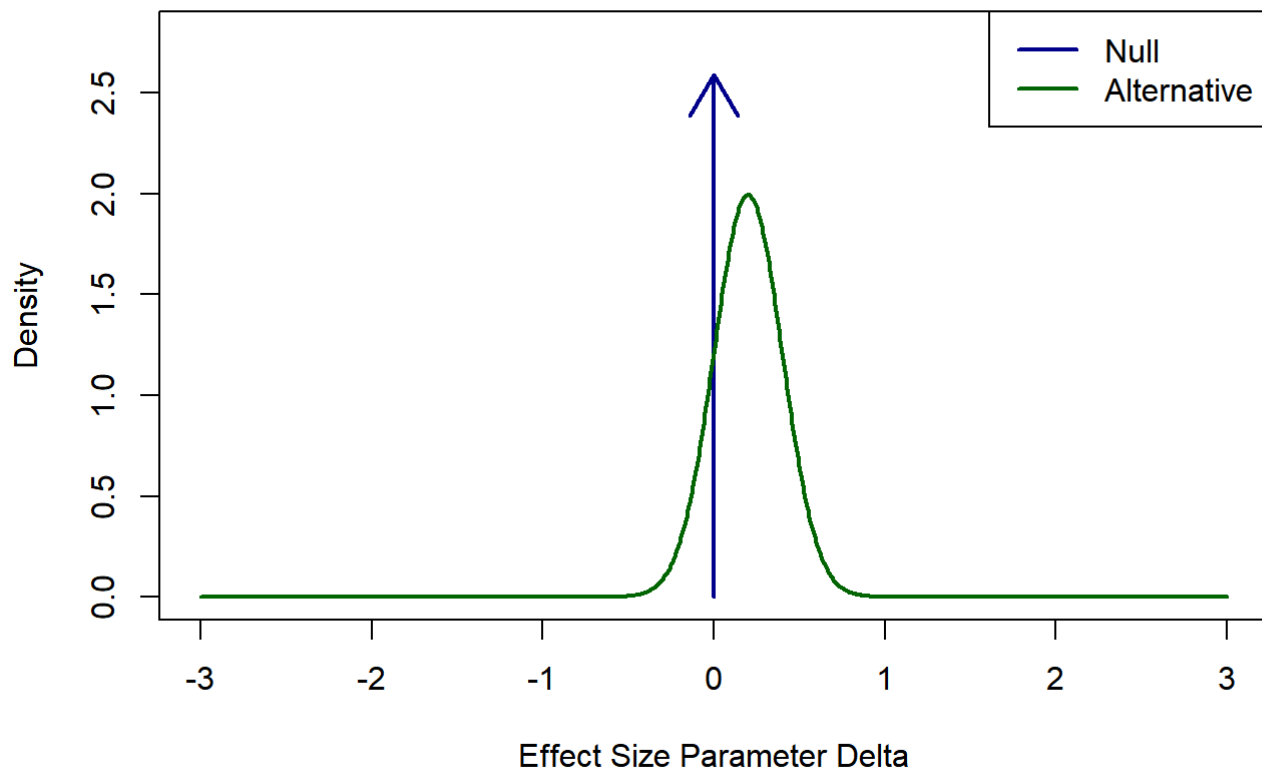
*plot(delta,f(delta),typ='n',xlab="Effect Size Parameter Delta",ylab="Density",ylim=c(0,1.4*maxAlt),main="Models")*

*arrows(0,0,0,1.3*maxAlt,col='darkblue',lwd=2)*

lines(delta,f(delta),col='darkgreen',lwd=2)

legend("topright",legend=c("Null","Alternative"),col=c('darkblue','darkgreen'),lwd=2)

Models



```

#dev.off()

#Prediction Function Under Null
nullPredF=function(obs,N) dt(sqrt(N)*obs,N-1)

#Prediction Function Under the Alternative
altPredIntegrand=function(delta,obs,N)
  dt(sqrt(N)*obs,N-1,sqrt(N)*delta)*f(delta)
altPredF=function(obs,N)
  integrate(altPredIntegrand,lower=lo,upper=hi,obs=obs,N=N)$value

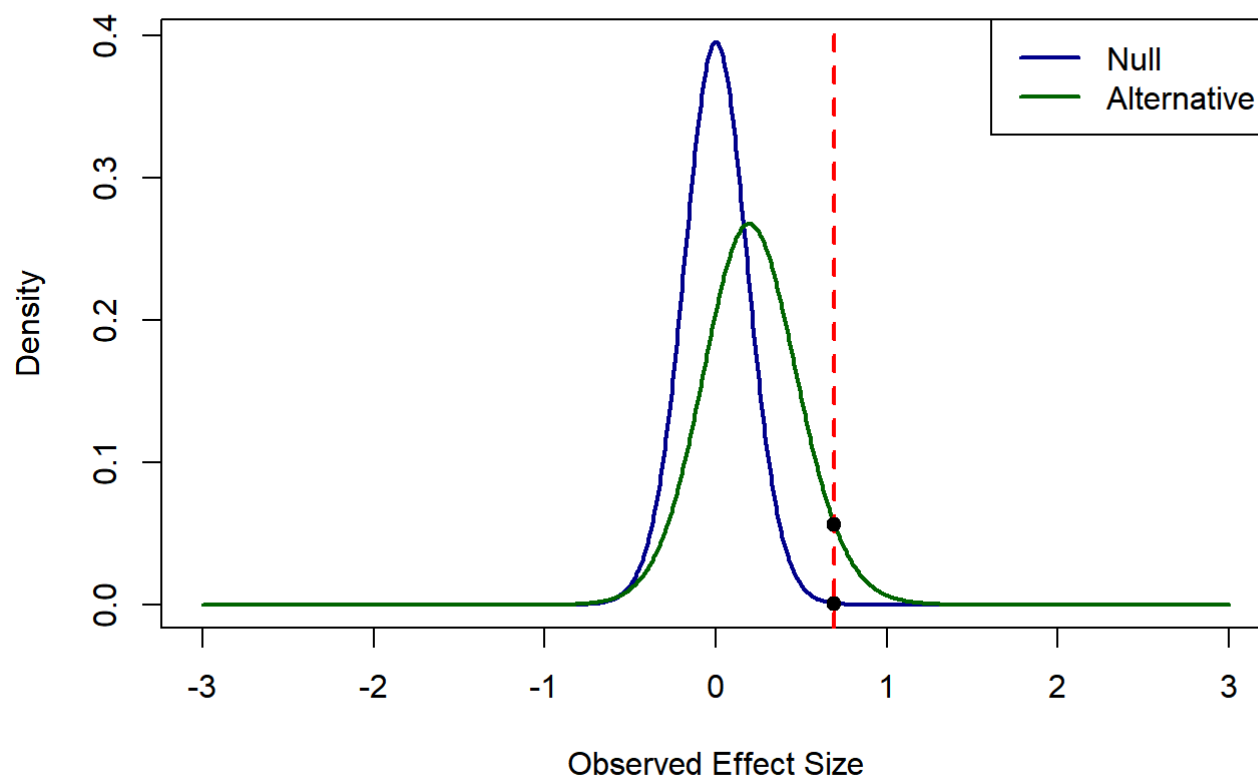
obs=delta
I=length(obs)
nullPred=nullPredF(obs,N)
altPred=1:I
for (i in 1:I) altPred[i]=altPredF(obs[i],N)

#Evaluate Predicted Density at Observed Value dz
valNull=nullPredF(dz,N)
valAlt=altPredF(dz,N)

#Plot The Predictions
#png(file=paste('posterior.png'),width=6000,height=4000, res = 1000)
top=max(altPred,nullPred)
plot(type='l',obs,nullPred,ylim=c(0,top),xlab="Observed Effect
Size",ylab="Density",main=paste("Bayes factor (alt/null) is ",round(valAlt/valNull,digits =3)),c
ol='darkblue',lwd=2)
lines(obs,altPred,col='darkgreen',lwd=2)
legend("topright",legend=c("Null","Alternative"),col=c('darkblue','darkgreen'),lwd=2)
abline(v=dz,lty=2,lwd=2,col='red')
points(pch=19,c(dz,dz),c(valNull,valAlt))

```

Bayes factor (alt/null) is 50.178



```
#dev.off()
```

```
cat("Bayes factor (alt/null) is ",valAlt/valNull," the t-value is ",sqrt(N)*dz," and the p-value is",2*(1-pt(abs(sqrt(N)*dz),N-1)))
```

```
## Bayes factor (alt/null) is 50.17844 , the t-value is 3.733364 and the p-value is 0.0008548068
```

```
#© Daniel Lakens, 2016.
```

```
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```